

# Mapping Loyalties

## *Visualizing EuroVision Voting Patterns Across Time*

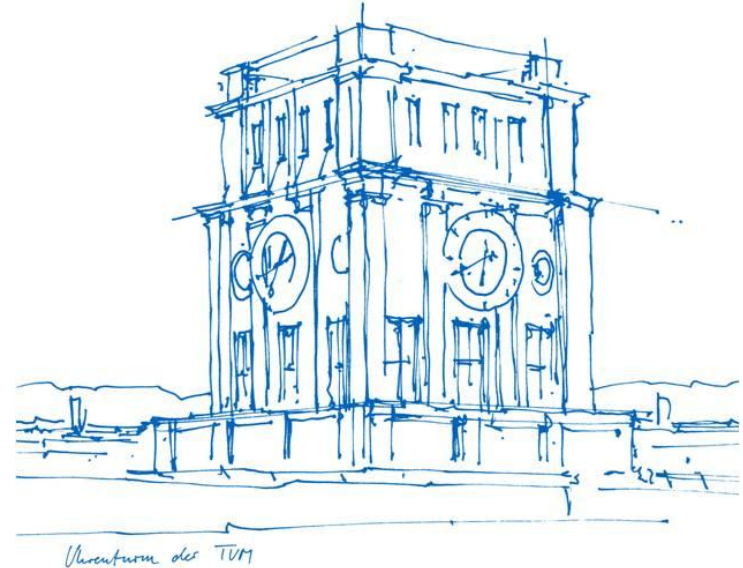
Supratik Patel, Marc Tagne

Technical University of Munich

Chair of Efficient Algorithms, Information, and Technology

Data Visualization Final Presentation

Heilbronn, 08.07.2026



# Introduction

**Eurovision has been running since 1957  
52 countries, 75K vote records,  
2 separate voting channels since 2016**



# The Problem Statement



## Three Questions addressed in the project:

- Which country pairs always vote for each other?
- Where do the juries and the public disagree?
  - Which relationships changed vote over time?



# Our Approach

## Four coordinated views, one shared state:

Every control (year range, vote type, region filter) updates all views at once.

No page reloads. No backend server.

Runs from a single folder, only used HTML CSS JS

If any changes to the data is needed, “npm run build” again

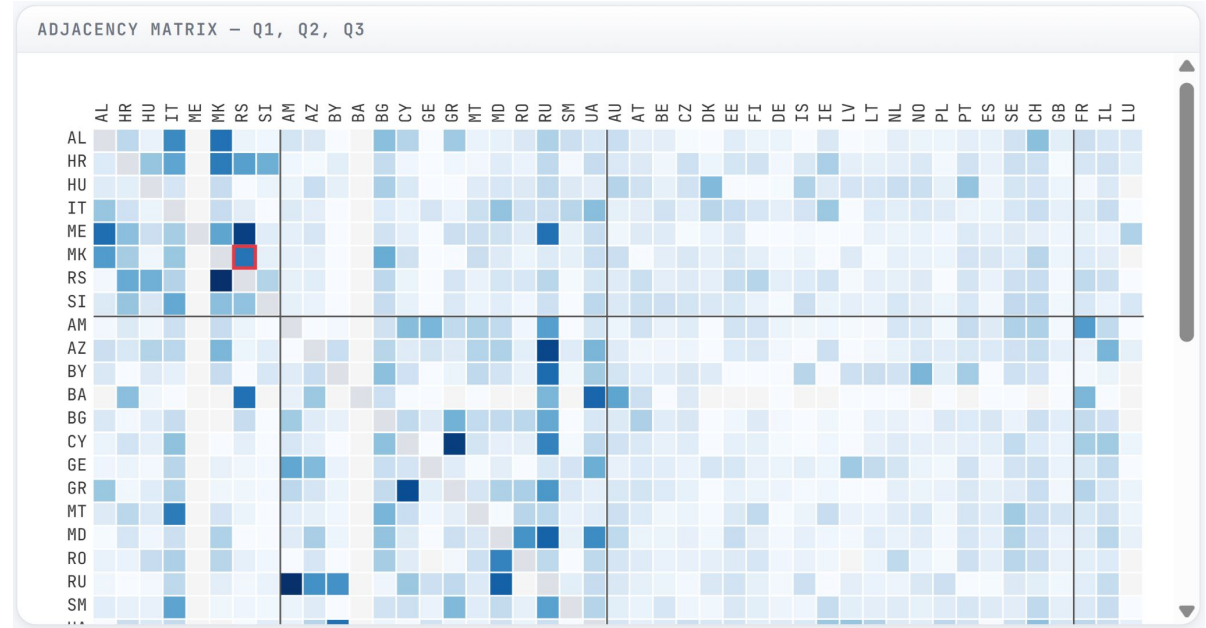




# View 1: Adjacency Matrix

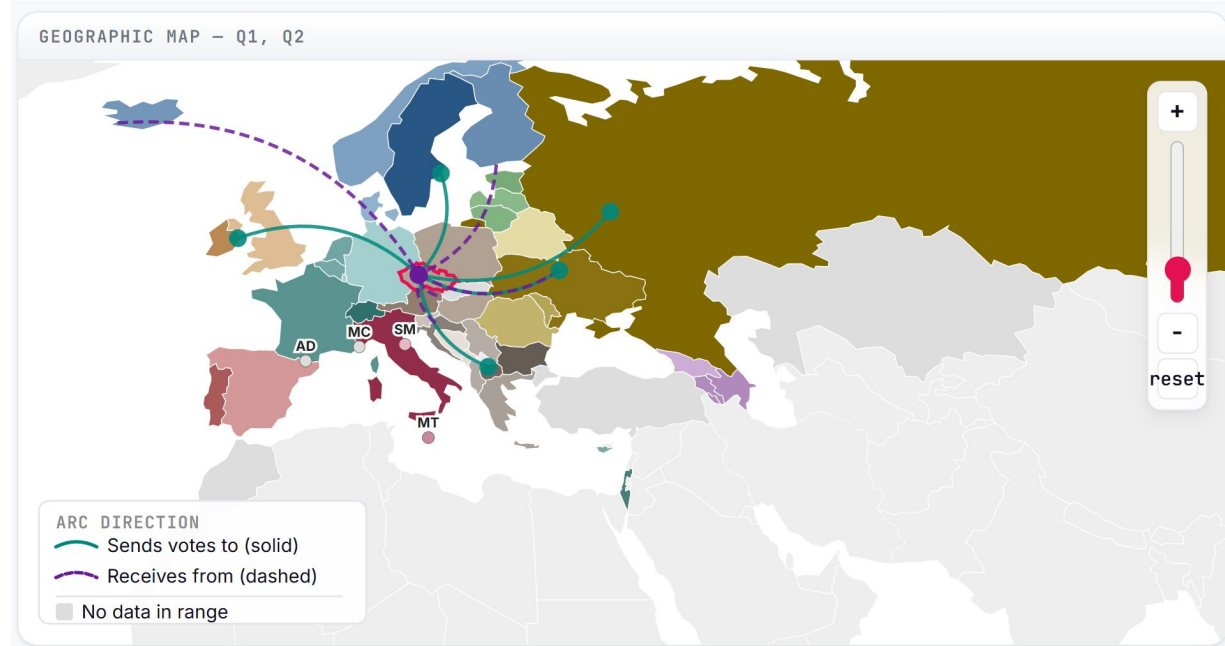
The main analytical view.

- Rows = voters,
- Columns = recipients
- Color = mean normalized support
- Sort by cluster, region, alphabetical or vote totals
- Click a cell to drill into the timeline



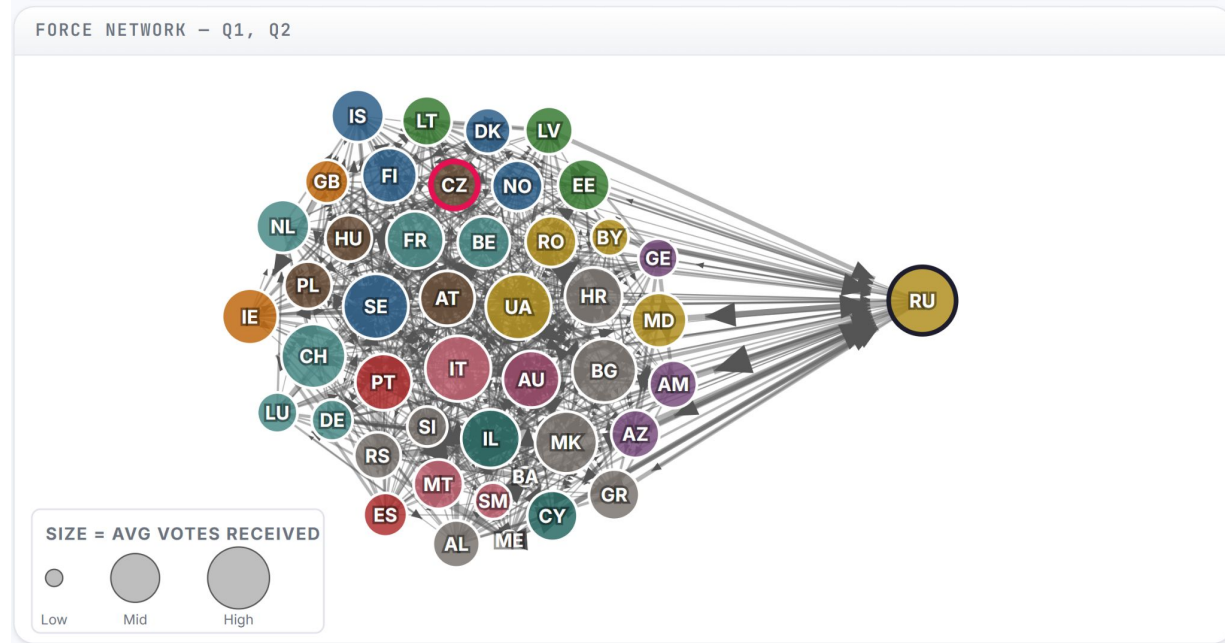
## View 2: Geographic Map

- **Bivariate color:** color family = region, brightness = avg. votes received
- **Select a country to see who it votes for (teal solid) and who votes for it (purple dashed)**
- **Mutual pairs curve in opposite directions so both arcs stay visible**
- **Zoom slider, micro-state pins, counter-scaled elements at any zoom level**



## View 3: Force Network

- Nodes sized by avg. votes received, colored by region
- Edge threshold slider (min. support) cuts the hairball down to readable
- Soft geographic anchoring: clusters echo the map layout (almost)

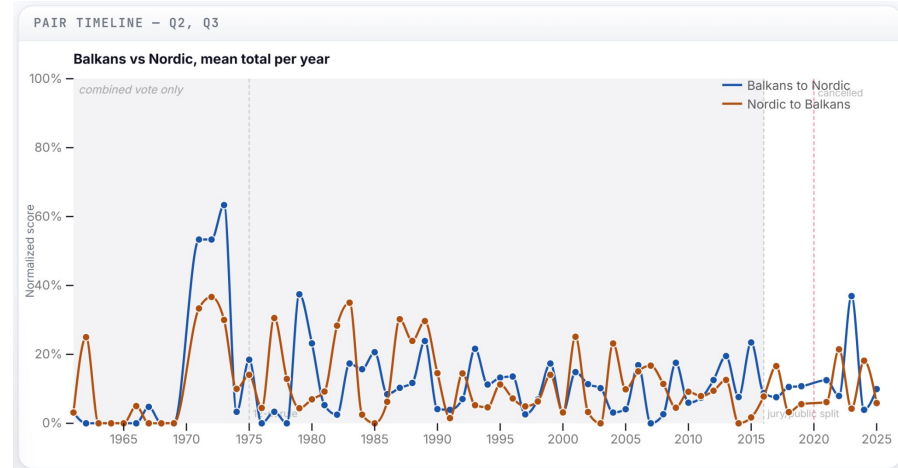
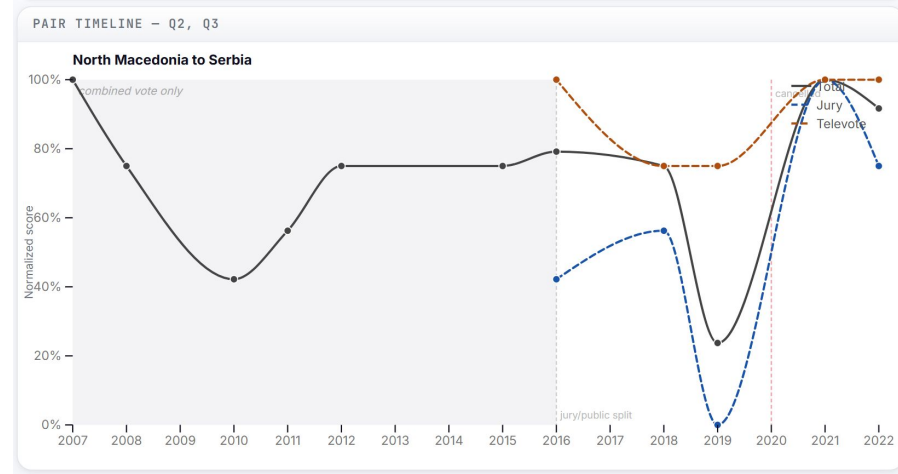




# View 4: Pair Timeline

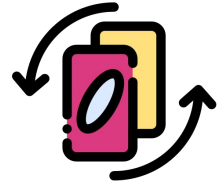
## Two modes:

- **Single pair: jury, public and total scores for one directed pair across all years.**  
Annotations mark the **1975(12pts. rule)**, **2016 (jury/public split)** and **2020 (cancelled)**.
- **Region comparison: pick two regions, see mean votes A to B and B to A.**  
e.g- do Nordics vote for Balkans as much as Balkans vote for Nordics?



## What we found

- **Q1. The strongest sustained pair in 2016-2025 is Cyprus to Greece at 95% over 6yrs, followed by Greece to Cyprus at 89% over 7yrs - the most dominant mutual pair.**
- **Q2. Bosnia to Ukraine had the biggest shift from 15.6% to 79.2%; i.e a 63.6% shift, from pre 2016 to modern era**
- **Q3. Slovenia to Croatia shows an 81% jury/public gap over 4 years (jury 8%, public 89%), which is the biggest divergence**



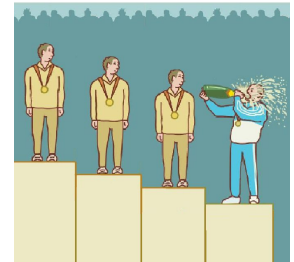
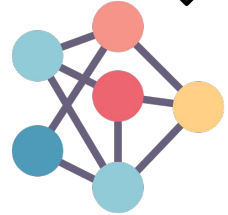
# Four Voting Blocs

Louvain community detection on the 2016–2025 total-vote graph found four clusters .The clusters do not match geographic regions; Italy is in the Balkan cluster, not Western Europe. France is isolated. These are voting communities, not cultural ones.

| Cluster                          | Countries                                                   | Why grouped                                                                                           |
|----------------------------------|-------------------------------------------------------------|-------------------------------------------------------------------------------------------------------|
| Western Balkans + Italy          | AL HR HU IT ME MK RS SI                                     | Heavy mutual voting; Italy and Hungary aligned with this bloc by their vote patterns, not geography   |
| Eastern network                  | AM AZ BA BG BY CY GE GR MD MT RO RU SM UA                   | Caucasus + eastern Europe + Cyprus/Greece; strong Caucasus mutual ties and Russia/Ukraine connections |
| Western Europe + Nordic + Baltic | AT AU BE CH CZ DE DK EE ES FI GB IE IS LT LV NL NO PL PT SE | Largest cluster; evenly distributed votes across the group                                            |
| Isolated                         | FR IL LU                                                    | No dominant voting partner; scatter across all other blocs                                            |

## Challenges and Fixes

- **France clicking Spain** : France's TopoJSON polygon includes French Guinea, pulling the geometric centroid into the Atlantic. Fixed by always using capital city coordinates for arc positions.
- **Zoom breaking micro-state pins** : Andorra and Monaco inflated at high zoom and blocked country clicks. Fixed by counter-scaling pins by  $1/k$  on every zoom event.
- **Cluster colors clashing with the map** : original single-color choropleth made region patterns invisible. Replaced with bivariate encoding (color family = region, brightness= score).
- **One-year flukes dominating rankings** : pairs with 100% mean support from a single year swamped Q1 and Q3. Fixed with minimum years filters (6 years for Q1, 4 for Q3) in the insights panel.



## Future Plans

- Better animations , learning and using modern libraries to make it more interactive
- Website deployed :- <https://github.com/SUP1MU/DataViz>
- More user evaluation and feedback with both Eurovision-familiar and unfamiliar participants

Thank you!